Interrupt

An **interrupt** is a signal that tells the CPU:

“Stop what you are doing, handle this important event first!”

Types of Interrupts

1. **Hardware Interrupts**
   Generated by external devices
   Example: Keyboard key press, mouse click.
2. **Software Interrupts**
   Generated by instructions inside a program
   Example: INT 21h (used in DOS).

DOS Interrupts

Purpose: Works with operating system services (printing, keyboard, program termination)

Works at: DOS/Operating System level

Useful INT 21h Functions

Function AH	Operation
01h	Read a character
02h	Print a character
09h	Print a string ($ terminated)
4Ch	Exit program

Example: Read a Key

```
MOV AH, 01h
INT 21h
; Key is now in AL
```

Example: Print a Character

```
MOV DL, 'A'
MOV AH, 02h
INT 21h
```

Example: Carriage Return

```
MOV DL,0DH
MOV AH, 02h
INT 21H
```

Example: New Line Insert

```
MOV DL,0AH
MOV AH, 02h
INT 21H
```

Example: Print a Sting

```
.MODEL SMALL
.STACK 100h
.DATA
msg DB "Hello World!$"   ; MUST end with $

.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    MOV AH, 09h           ; print-string function
    MOV DX, OFFSET msg    ; string address
    INT 21h               ; call DOS interrupt
    MOV AH, 4Ch           ; exit
    INT 21h
MAIN ENDP
END MAIN
```

Example: Terminate or Exit Program

```
MOV AH, 4Ch          ; exit
INT 21h
```

BIOS Interrupts

Purpose: Works with **screen/graphics** (printing text, cursor, colours, scrolling).

Works at: Hardware/BIOS level

Useful INT 10h Functions

Function AH	Operation
02h	Set cursor Position
03h	Get cursor Position
06h	Scroll Up
09h	Write Character & Attribute
0Eh	Teletype Output Function

Example: Set Cursor Position

```
MOV AH, 02h
MOV BH, 00h
MOV DH, 5     ; row
MOV DL, 10    ; column
INT 10h
```

Example: Get Cursor Position

```
MOV AH, 03h
MOV BH, 00h
INT 10h
```

Example: Scroll Up

```
MOV AH, 06h
MOV AL, 01h      ; scroll 1 line
MOV BH, 07h      ; attribute of blank (white on black)
MOV CX, 0000h    ; upper-left (0,0)
MOV DX, 184Fh    ; lower-right (24,79)
INT 10h
```

Example: Write Character & Attribute

```
MOV AH, 09h
MOV AL, 'A'
MOV BH, 00h
MOV BL, 0Eh     ; yellow on black
MOV CX, 1
INT 10h
```

Example: Teletype Output Function

```
MOV AH, 0Eh       ; teletype output
MOV BH, 0         ; page 0
MOV BL, 0Eh       ; yellow color
INT 10h
```

BIOS Keyboard Services

Purpose: Works with keyboard input (reading keys)

Works at: Hardware/BIOS level

Useful INT 16h Functions

AH	Operation	Detail
00h	Read key (wait for key press)	CPU waits until user presses a key. Returns ASCII in AL, scan code in AH
01h	Check if key is pressed (no wait)	CPU does not wait. ZF = 1 if no key pressed, ZF = 0 if key pressed. AL = ASCII, AH = scan code

Example: Wait for key

```
MOV AH, 00h
INT 16h          ; Wait for key
; AL = ASCII of key
; AH = scan code
```

Example: Check key

```
MOV AH, 01h
INT 16h          ; Check key
JZ NO_KEY        ; Jump if no key pressed (ZF=1)
; If key pressed:
; AL = ASCII
; AH = scan code
```

Complete code of 16h

```
.MODEL SMALL
.STACK 100h
.DATA
msg DB "Press a key: $"
.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    ; Print message using DOS (INT 21h, AH=09h)
    MOV DX, OFFSET msg
    MOV AH, 09h
    INT 21h

WAIT_KEY:
    MOV AH, 01h      ; Check if key pressed
    INT 16h
    JZ WAIT_KEY      ; Loop until key pressed

    ; Key pressed, read it
```

```
    MOV AH, 00h
    INT 16h          ; AL = key ASCII

    ; Print key using teletype
    MOV AH, 0Eh
    MOV BH, 0
    MOV BL, 0Eh      ; yellow
    INT 10h

    ; Exit
    MOV AH, 4Ch
    INT 21h
MAIN ENDP
END MAIN
```